

Manojkumar Bontha

Data Engineer / AWS / Azure / Python Developer

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Professional Summary

- Having 5+ years of experience. Big Data Environment, Hadoop Ecosystem, and AWS. Interacting with business users to analyze the business process and requirements and transforming requirements into data warehouse design, documenting, and rolling out the deliverables.
- Strong working experience in Cloud data migration using AWS and Snowflake
- Experience in a Data Integration platform like AWS Glue.
- Extensive knowledge of Spark Streaming, Spark SQL, and other Spark components such as accumulators, broadcast variables, various levels of caching, and optimization techniques for Spark employment
- Hands-on experience with Bigdata ecosystem implementation, including Hadoop MapReduce, NoSQL, Informatica cloud services, Fivetran, Apache Spark, Pyspark, Python, Scala, Hive, Impala, Sqoop, Kafka, AWS, Azure, and Oozie.
- Defined product requirements and created high level architectural specifications to ensure that existing platforms are feasible and functional.
- Prototyped components were benchmarked, and templates were provided for development teams to test design solutions.
- Familiar with data processing performance optimization techniques such as dynamic partitioning, bucketing, file compression, and cache management in Hive, Impala, and Spark
- Experience with various data formats such as Json, Avro, parquet, RC and ORC formats and compressions like snappy & bzip.
- Working knowledge in AWS environment and AWS spark, Snowflake, Lambda, AWS RedShift, DMS, EMR, RDS, EC2, AWS stack with Strong experience in Cloud computing platforms such as AWS services.
- Migrated an existing on-premises application to AWS. Used AWS services like EC2 and S3 for small data sets processing and storage, experienced in Maintaining the Hadoop cluster on AWS EMR.
- Experience in Data Pipelines, phases of ETL, ELT data process, converting Big Data/unstructured data sets (JSON, log data) to structured data sets for Product analysts, Data Scientists.
- As a Data Engineer, responsible for data modeling, data migration, design, and ETL pipeline preparation for both cloud and Exadata platforms.
- Agile Program management, change management, User management JIRA-Agile Scrum/Kanban, Bug Cycle management, SDLC design advice using JIRA.
- Experience with Atlassian JIRA installation, administration, and maintenance.
- Extensive experience in integrating various data sources like SQL Server, DB2, PostgreSQL, Oracle, and Excel.
- Strong Data Warehousing knowledge with Informatica, including considerable experience creating Tasks, Workflows, Mappings, Mapplets, and Scheduling Workflows and Sessions.
- Experienced in using object-oriented programming (OOP) concepts using Python.
- Solid knowledge of usability engineering, user interface design and development.
- Outstanding knowledge of Reporting tools like Power BI, Data Studio, and Tableau.
- Good backend skills like creating SQL objects like tables, Stored Procedures, Triggers, Indexes and Views to facilitate Data manipulation and consistency.
- Expertise in leveraging and implementing the best SDLC and ITIL techniques.
- Team handling experience, which includes work planning, allocation, tracking and execution. Relationship driven, result driven and Creative out of the box thinking.
- Proficient in creating and maintaining data models in Power BI, employing star and snowflake schemas, managing relationships, and establishing hierarchies.
- Experienced in data warehousing technologies, including SQL Server Analysis Services (SSAS), Azure Analysis Services, and Amazon Redshift.
- Familiarity with various BI tools and platforms, such as Tableau, Qlik Sense, and Microsoft Reporting Services (SSRS).

- Expertise in writing DAX expressions for calculating measures, creating calculated columns, and performing data transformations.
- Skilled in developing interactive and informative data visualizations using Power BI, encompassing charts, maps, and tables.
- Proficient in utilizing Power Query for data cleaning, transformation, and preparation.
- Capable of writing and executing SQL queries to extract data from relational databases.
- Experience with programming languages such as Python, R, and M for automating data tasks and custom BI solutions.
- Familiarity with cloud-based BI platforms, including Power BI Services and Azure Synapse Analytics.

Technical Skills

- **Big Data Technologies:** Hadoop, MapReduce, HDFS, Sqoop, PIG, Hive, HBase, Oozie, Flume, NiFi, Kafka, Zookeeper, Yarn, Apache Spark, Mahout, Sparklib
- **Databases:** MySQL, SQL Server, MongoDB, Cassandra, DynamoDB, PostgreSQL, Oracle, Teradata, Cosmos.
- **Programming:** Python, R, PySpark, Scala, Java, C, C++, Shell script, Perl script, SQL
- **Cloud Technologies:** Microsoft Azure, AWS EC2, S3, Glue, Batch, ECS Fargate, Dynamo DB, EMR, RDS, ELB, Lambda, SNS, SQS, Redshift, Cloud Front, and CloudWatch.
- **Frameworks:** Django REST framework, MVC, Hortonworks
- **Tools:** PyCharm, Eclipse, Visual Studio, SQL*Plus, SQL Developer, TOAD, SQL Navigator, Query Analyzer, SQL Server Management Studio, SQL Assistance, Eclipse, Postman
- **Versioning tools:** SVN, Git, GitHub
- **Operating Systems:** Windows 7/8/XP/2008/2012, Ubuntu Linux, MacOS
- **Network Security:** Kerberos
- **Database Modelling:** Dimension Modeling, ER Modeling, Star Schema Modeling, Snowflake Modeling
- **Monitoring Tool:** Apache Airflow
- **Visualization/ Reporting:** Tableau, ggplot2, matplotlib, SSRS and Power BI
- **Machine Learning Techniques:** Linear & Logistic, Regression, Classification and Regression Trees, Random Forest, Associative rules, NLP, and Clustering.

Professional Experience

DCFS – SPRINGFIELD, ILLINOIS

01/2023 to 12/2023

Big Data Engineer

Responsibilities:

- Actively involved in designing, developing, and analyzing data analytics solutions using Power BI, Azure Data Factory, and Azure SQL Server.
- Built data pipelines in Azure Data Factory to integrate data from on-premise sources.
- Implemented incremental refresh in Power BI for combined analytics and publishing to Power BI Service.
- Utilized DAX for creating calculated columns and measures in Power BI.
- Conducted data modeling in Power BI to identify entities, properties, and relationships.
- Analyzed event registration data using Power BI and SQL Server to derive insights into attendee demographics, attendance trends, and ticket sales. Applied insights to optimize event logistics.
- Developed a Power BI dashboard for tracking and analyzing event performance data, improving marketing campaigns and outreach efforts.
- Identified attendee segments with similar interests and needs using Power BI and SQL Server, optimizing event content and activities.
- Implemented row-level security in Power BI reports using DAX.
- Developed Paginated Reports and implemented Drill Down and Drill Through in Power BI Reports.
- Leveraged external tools Bravo & Dax Studio for DAX queries and performance tuning.
- Manage and administer Teradata and Vertica database environments.
- Develop and maintain data models and schemas that facilitate efficient data processing.
- Implement data ingestion processes to bring data from various sources into the big data ecosystem.

- Integrate and transform diverse data sets to make them compatible for analysis.
- Develop and optimize distributed processing systems using tools like Apache Hadoop, Apache Spark, or other big data frameworks.
- Implemented batch and real-time data processing pipelines.
- Choose and implement appropriate data storage solutions, including distributed databases (e.g., HBase, Cassandra) and distributed file systems (e.g., HDFS).
- Optimized data storage and retrieval for performance and scalability by 30%.
- Implement data quality checks and validation processes to ensure accuracy and reliability of data.
- Work on data governance initiatives to establish standards and policies for data management.
- Implement security measures to protect sensitive data and ensure compliance with privacy regulations.
- Work on encryption, access controls, and data anonymization strategies.
- Optimize big data solutions for scalability, performance, and efficiency.
- Knowledge of performance tuning strategies for Hive and Trino.

AMAZON – HYDERABAD, IN
DATA ENGINEER

05/2020 to 08/2022

Responsibilities:

- Design robust, reusable, and scalable data driven solutions and data pipeline frameworks to automate the ingestion, processing and delivery of both structured and unstructured batch and real-time streaming data using Python Programming.
- Design and develop ETL Data Pipelines to migrate Abinitio jobs to Python using Pyspark.
- Develop Lambda functions in python using AWS Lambda, Kinesis and Elastic Search which invokes python scripts to perform various transformations and analytics on large data sets in AMAZON EMR.
- Iterate rapidly and work collaboratively with product owners, developers, and other members of the development team.
- Defect tracking and resolving bug-fixes, hot-fixes by merging the feature branch code to the master branch in GIT source control management.
- Use Amazon Elastic Cloud Compute (EC2) infrastructure for computational tasks and Simple Storage Service (S3) as Storage mechanism.
- Work with different types of data sources such as application data, process data and formats like CSV, JSON, Parquet.
- Querying multiple databases like PostgreSQL, Oracle SQL and Teradata for data-processing.
- Automated configuration of monitoring tools to enhance our awareness of the availability and utilization of AWS resources.
- Maintain AWS Data Pipeline as web service to process and move data between Amazon S3, Amazon EMR and Amazon RDS at specified intervals.
- Using Python class approach for wrapping the script and deploy into production environment.
- Troubleshoot and resolve data processing issues and proactively engaged in data modelling discussions.
- Write programs in Spark using Python, Pyspark packages for performance tuning, optimization and data quality validations.
- Work closely with Teams across the BU to create comprehensive test tools and automation frameworks.

WIPRO – HYDERABAD, INDIA
DATA ENGINEER

05/2019 to 05/2020

Responsibilities:

- Involved in reviewing business requirements and analyzing data sources from Excel, SQL Server for design, development, testing, and production rollover of reporting and analysis projects.
- Created ETL packages dealing with different data sources (SQL Server, Flat Files) and loaded the data into target data sources by performing different kinds of transformations using SQL Server Integration Services (SSIS)

- Created views in Tableau Desktop that were published to internal team for review and further data analysis and customization using filters and actions.
- Collaborated with data engineering team and data operations team in smoothening ETL pipeline and used Python and SQL to automate report delivery and make continuous enhancement.
- Analyzed the applications and created the test data required for functional testing and regression testing.
- Managed documented and maintained test cases, test scripts and test results in Quality Center and provided status update report using MS Excel
- Developed Dashboards using calculations, parameters in Tableau and End to end experience in designing and deploying data visualizations using Tableau.
- Created various reports like tabular, matrix, list, parameterized, linked and other reports using SSRS and provide data validation through SSRS reports to end users on monthly/Ad hoc basis.
- Created Context filters on the existing filters and Tableau data extract to publish it in server by optimizing it to reduce the report rendering time, used performance actions while handling huge volume of data.
- Recommended new methods, tools and shared best practices with internal BI team to streamline reporting processes and enable end user self-serve data capabilities.
- Extensive experience in developing end-to-end orchestration layer for data pipelines using frameworks like Apache Airflow, Prefect, Databricks Workflow.

ASCII - HYDERABAD, INDIA
DATA ENGINEER

02/2018 to 04/2019

Responsibilities:

- Worked with SSRS for creating and configuring dashboards, graphs, and reports for analysis from different database sources like SQL Server, Excel, CSV, and Text files.
- Documented processing times for each module, developed test cases and used them to run through each process, tuned the matching parameters based on test results.
- Knowledge of statistics and experience using statistical packages for analyzing datasets (Excel, SPSS, SAS etc.)
- Strong knowledge and experience with reporting packages (Business Objects etc.), databases (SQL etc.), programming (XML, JavaScript, or ETL frameworks)
- Developed and implemented a predictive modeling framework that increased sales revenue by 30% through accurate forecasting of market trends.
- Created and maintained a comprehensive data governance program that ensured compliance with industry regulations and improved data quality by 25%.
- Implemented a reporting and analytics platform resulting in a 20% increase in data-driven decision making across the organization.
- Developed and deployed machine learning algorithms to optimize pricing strategies, resulting in a 20% increase in profit margins.
- Design and implementation of a data lake infrastructure, resulting in a 40% reduction in data storage costs and a 25% increase in data accessibility.
- Led a team of data analysts in the design and execution of a data warehousing project, resulting in a 50% reduction in data processing time and a 20% improvement in data accuracy.

Education:

Masters in Mathematics and Computer Science

University Of Illinois Springfield • Springfield, ILLINOIS, USA • 08/2022 - 12/2023

Bachelor of Technology in Computer Science

Vardhaman College Of Engineering • Hyderabad, Telangana, INDIA • 08/2014 - 08/2018